DAVID NARAY

UofG Computer Science Graduate (BSc Hons) — Software Development & Systems
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WHAT I DELIVER

Computer Science graduate from the University of Glasgow with experience developing software in Python, C, Java and JavaScript. Projects include real-time multiplayer systems, multithreaded applications, neural networks, image processing and full-stack web applications. Familiar with networking, automated testing, CI/CD and Agile development, with practical experience supporting live software and server infrastructure at Orkora.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, SQL, HTML/CSS
Web & Backend: Node.js, Django, WebSockets, AJAX, MongoDB, relational databases, form and file-upload handling, client–server and server-authoritative design
AI, Graphics & Vision: Neural networks (backpropagation, gradient descent), genetic algorithms, OpenCV, MediaPipe, NumPy, image-processing kernels, Three.js (GPU instancing), Canvas, PyQt5, Blender Python scripting

Systems & Infrastructure: Multithreading (pthreads), pipes and inter-process communication, TCP sockets, Docker, DNS and HTTPS (Cloudflare), static hosting, server hardware, rack cabling and UPS fault diagnosis

Algorithms: A* and hierarchical pathfinding, priority scheduling, data smoothing, geometric heuristics
Practices: Unit testing (enforced code-coverage threshold), CI/CD, release engineering, Agile/Scrum with a real client
Tools: Git, GitHub, GitHub Pages, Docker, Maven, Cloudflare, Blender

PROJECTS

Hand Gesture Control for 3D Software (Dissertation) — Control Blender with hand signs in front of a webcam instead of a mouse. Hand landmarks from MediaPipe are smoothed and classified with geometric rules, then sent as commands over TCP to a Python script running inside Blender. The tracking side runs in a Docker container fed by the webcam over a local port, and a PyQt5 overlay draws a live cursor dot for each hand.
Tech: Python, MediaPipe, OpenCV, NumPy, PyQt5, Docker, TCP sockets **|** **Key concepts:** Data smoothing, geometric heuristics, inter-process communication, code environments

Multiplayer 3D City Builder (Browser) — Players build a city together on procedurally generated 3D terrain, and the units find their own routes. The server decides what is true in the game, so every player sees the same world, and pathfinding stays fast on large maps by planning in two levels.
Tech: JavaScript, Node.js, Three.js, WebSockets, Python **|** **Key concepts:** A* & hierarchical pathfinding, GPU instancing, server-authoritative multiplayer

Client Project (University, 5-person Agile team) — Built software for a real client in a five-person Scrum team, working in sprints and directly with the client. Automated tests ran in the CI/CD pipeline and had to meet an enforced code-coverage threshold.
Tech: Java, Maven, CI/CD **|** **Key concepts:** Agile/Scrum, enforced test coverage, working with a client

Multithreaded Task Scheduler (C) — A priority task queue that runs jobs in parallel, so urgent work starts first and waiting time drops. Jobs queue by priority and run across threads, with pipes between processes.
Tech: C, pthreads, Pipes (IPC) **|** **Key concepts:** Concurrency, priority scheduling

AI Racecars — Simulated cars learn to drive faster laps over successive generations. A fitness function scores each run, and the best neural-network drivers are carried forward.
Tech: JavaScript, canvas **|** **Key concepts:** Fitness functions, genetic AI, neural networks

Line Fitting — 4 points the user can move provide targets for a neural network to pass a line through. The input being x, the output y, measuring loss by the difference of expectation and reality.
Tech: JavaScript, canvas **|** **Key concepts:** Fitness functions, backpropagation, gradient descent, neural networks

Filter — vary the size of an area (kernel) that applies a transform on pixel data to achieve three different effects: edge detection, sharpening and blurring.
Tech: JavaScript, canvas, forms **|** **Key concepts:** kernels, image processing, computer vision

Web Platform — A full-stack web app where forms and file uploads update without reloading the page. Django handles the backend, and AJAX requests keep the interface responsive.
Tech: Python, Django, JavaScript, AJAX **|** **Key concepts:** Asynchronous requests, full-stack development

Personal Website — My portfolio site at davidnaray.dev: a multi-page static site hosted on GitHub Pages. I set up the custom .dev domain and its DNS records through Cloudflare.
Tech: JavaScript, Git, GitHub Pages, Cloudflare DNS **|** **Key concepts:** Static hosting, DNS, HTTPS

EXPERIENCE

Contract Programmer, Orkora Ltd. Aug – Sep 2026

- Managed machinery-tracking software, keeping it running reliably.
- Consolidated two server racks into one, rewiring and reorganising cabling.
- Diagnosed a faulty UPS (battery backup) that would not charge, identifying why servers were down.

Kitchen Porter, Cults Hotel Jan – Aug 2022

Kitchen Assistant, Codonas Jun – Aug 2021

Volunteer, Barnardos Mar – Nov 2019

EDUCATION

BSc (Hons) Computer Science, University of Glasgow 2020 – 2026
Relevant modules: Systems Programming (C, concurrency), Algorithms (graph search, complexity), Software Release Engineering (testing, CI/CD), Database Systems (SQL), Artificial Intelligence, Web App Development, Robotics Foundations, Text as Data (text analysis), Data Fundamentals

ADDITIONAL

Languages: English (Native)
Awards: STEM Silver Award, Duke of Edinburgh (Bronze), Grade 8 Piano & Saxophone, First Aid Training
Activities: Air Cadets (3 years)